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Course: Data Wrangling

Project Report: Airbnb Market Insights

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Contribution of each project Members:

Roll No.	Name:	Contribution
L001	Srishti	Data extraction, cleaning, and standardization; cleaning log and documentation; machine learning model development and evaluation; written report
L021	Yash Hotey	Power BI dashboard design and build; dashboard data export; GitHub repository management
L025	Akshada Jangam	Data transformation and feature engineering; reviews and calendar aggregation; SQLite schema design and loading; insight generation and EDA visualizations; presentation deck

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1. Problem Statement

Airbnb pricing and host performance vary considerably across cities, neighbourhoods and listing types, but the drivers are unclear. Inconsistent raw data, assumptions about price and host quality, and the perceived dominance of location make cross-city analysis difficult. The project addresses three problems:

- Data inconsistency: Inside Airbnb data has missing values, inconsistent formats and city-specific differences; a standardised, analysis-ready dataset is needed.
- Price vs host performance: Superhost status reflects reliability and service quality, not price or demand. Are they related or independent measures of success?
- Listing characteristics vs location: Neighbourhood is widely seen as a major price driver, but its influence relative to capacity and other features, and its consistency across cities, is unclear.

2. Data Extraction

Data came from Inside Airbnb (insideairbnb.com), a public open dataset (CC BY 4.0, free to use with attribution). No API, authentication or scraping was needed: 8 pre-scraped CSV files (listings, calendar, reviews, neighbourhoods for NYC and LA; scrape date 2026-08-11) were downloaded as ZIP archives. Provenance fields (scrape_id, last_scraped, source) were retained. About 29.9 million raw rows were ingested.

Raw data profile: Total raw rows ingested: ~29.9 million.

Table	NYC Rows	LA Rows	Columns
Listings	30,234	43,781	90
Calendar	11,035,410	16,035,198	5
Reviews	994,127	1,856,294	6
Neighbourhoods	230	270	2

Access & licensing:

- License: CC BY 4.0 - Inside Airbnb is free for commercial and non-commercial use with attribution
- Access constraints: None encountered - files are directly downloadable as ZIP archives per city
- Rate limits / authentication: N/A - data is pre-scraped and served statically
- Scrape provenance: Each listings.csv row includes scrape_id, last_scraped, and source fields that we retained during ingestion to track per-file provenance.

3. Data Cleaning

A single clean_listings() function was applied to both cities for consistent treatment; all 126 decisions (18 action types) are documented in cleaning_log.csv.

3.1. Cleaning

- Missing values (imputed): price was missing in 32.75% of NYC and 16.20% of LA listings (22.96% overall) and is median-imputed by city at modelling time; minimum_nights was median-filled; reviews_per_month was set to 0 when there were no reviews, else median-imputed; missing host info became “Not available”.
- Meaningful missingness (retained): missing review dates and review_scores_* indicate listings with no reviews. *_was_missing flags were created before imputation.
- Structural absence (dropped): bathrooms (100% missing in LA) and eleven empty columns.
- Duplicates and orphans: no duplicate listing IDs or full-row duplicates; 121,531 orphan reviews (4.26%) with no matching listing were removed.
- Type corrections: price stripped of $ and commas; dates parsed to datetime; “t/f” flags mapped to Boolean/0-1; 26 numeric columns coerced with pd.to_numeric.
- Standardisation: room types mapped to four categories; neighbourhood fields merged under a common name with a city_neighbourhood key (223 NYC and 265 LA neighbourhoods validated). A case-sensitivity bug (“Not Available” vs “Not available”) misclassified 30,740 LA licences as Registered; fixed so licences are Missing (55,755), Registered (11,609) or Exempt (6,802).
- Deferred: missing price (22.96%) and bedrooms/beds (32–41% NYC, 0–2% LA) are imputed at modelling time with flags kept.

4. Data Transformation

- Reviews aggregation: 2,850,421 reviews grouped by listing into reviews_total, reviews_last_90d, reviews_last_365d and review_recency_days (as_of 2026-08-20). 54,546 listings have reviews; 19,469 have none.
- Calendar aggregation: 27,070,608 rows collapsed per listing into cal_days, cal_booked_days, cal_occupancy_rate (booked ÷ total days), cal_min_min_nights and cal_max_max_nights; 69,152 listings matched (93.4%), 4,863 had no calendar data.
- Engineered features: host_tenure_months, host_portfolio_bucket, price_vs_city_median, reviews_per_host_year and is_professional_host (>5 listings).
- Outliers: price_vs_city_median capped at p99 (11.24; 586 listings); reviews_per_host_year capped at 365 (29 listings); price capped per city at p99 (NYC $1,676; LA $2,588) during modelling.

5. Data Loading and Storage

The data was stored in a SQLite database (airbnb.db, 216.4 MB) with two tables (listings: 74,015 rows; reviews: 2,728,890 rows), five indexes (listing ID, city, city_neighbourhood, reviews by listing, reviews by date) and enforced foreign keys. Integrity checks passed with zero foreign-key violations, and listing counts matched the sources (30,234 NYC; 43,781 LA). The 27M calendar rows were stored only as seven listing-level features, the ~500 MB free-text comments field was excluded, Booleans were stored as INTEGER (0/1/NULL), and a round-trip check confirmed the data matched the modelling DataFrame.

6. Insights and Communication

- Insight 1: LA hosts are superhosts ~1.8× more often than NYC hosts (38.6% vs 21.5%), the opposite of what the more competitive market would suggest.
- Insight 2: professionalisation does not explain the gap; LA's professional-host share (36.7%) is only ~3 points above NYC's (33.9%).
- Insight 3: LA has a wider luxury tail; its 95th-percentile price is 4.98× the city median vs 3.98× in NYC.
- Insight 4 (lead), the superhost paradox: median calendar occupancy is higher in NYC (39.2%) than LA (28.5%) despite LA's ~2× superhost rate. Superhost criteria (rating ≥4.8, ≥10 stays/year, <1% cancellation, ≥90% response rate) measure host conduct, not demand. Candidate explanations: the badge is a quality signal, not a demand signal; NYC supply is constrained by Local Law 18; and LA has ~45% more listings, diluting demand.

Limitations

- Snapshot data, not a time series, so seasonality and price changes cannot be observed. Scrape windows differ by ~8 weeks, so the occupancy gap is partly confounded by season.
- Occupancy is a demand proxy derived from host-controlled availability (it includes maintenance or personal blocks); price is advertised, not transacted; superhost status is a point-in-time snapshot of a rolling 12-month evaluation.
- Wrangling choices may bias results: bathrooms dropped, 22.96% of prices missing in EDA, price clipped at p99, and the >5-listing professional threshold is a working definition. Coverage bias is possible for new, hidden or hotel inventory.

7. Machine Learning

Two supervised tasks were each compared against a naive baseline and a more expressive model. Imputation, scaling and one-hot encoding were done inside sklearn Pipelines fit on training data only (no leakage); cross-validation was skipped due to Colab limits, so metrics

come from one held-out 20% test set (random_state=42). All six fitted pipelines were saved as .joblib files.

7.1. Task 1 — Regression: Predicting Nightly Price

Target log1p(price); 35 features (29 numeric, 6 categorical); 45,614 train / 11,404 test rows. price_vs_city_median was excluded as target-derived, and latitude/longitude were replaced by categorical neighbourhood features.

Model	RMSE (log)	MAE (log)	R^2	Fit time
Baseline (median predictor)	0.899	0.711	-0.003	<1s
Linear Regression	0.443	0.327	0.757	6.3s
Random Forest	0.399	0.287	0.802	178s

Random Forest gains only ~4.5 R^2 points over Linear Regression at 28× the training time, so price is largely linearly predictable. Top features: accommodates (0.376), bedrooms (0.106), cal_min_min_nights (0.090), room_type = Entire home/apt (0.063) and minimum_nights (0.043). Property size explains ~48% of variance, while location features contribute under 5% combined, so once size is controlled, cross-neighbourhood variation is modest.

7.2. Task 2 — Classification: Predicting Superhost Status

Target host_is_superhost (31.9% positive); 23 features (12 review-derived features dropped as leakage); 58,575 train / 14,644 test rows, stratified by city.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Fit time
Baseline (majority)	0.683	0.000	0.000	0.000	0.500	<1s
Logistic Regression	0.699	0.519	0.678	0.588	0.766	9.0s
Random Forest	0.786	0.625	0.813	0.707	0.873	13.2s

Random Forest (ROC-AUC 0.873, +11 points over Logistic Regression) shows superhost status involves strong nonlinear interactions. Top features: number_of_reviews (0.113), calculated_host_listings_count (0.090), reviews_total (0.083), cal_occupancy_rate (0.059)

and cal_booked_days (0.057). No location feature appears in the top 15, so superhost is a host-quality signal, not a location-driven trait.

Where the models underperform: Luxury listings above $1,500/night are systematically underpredicted (only 1% of training rows); “high-occupancy non-superhosts” are often misclassified; and LA’s wider luxury tail is only partly captured, with heteroscedastic residuals across cities.

8. Power BI Dashboard

The single-page “Airbnb Market Insights” dashboard (Power BI Desktop, published to Power BI Service) uses listings_clean.csv (74,015 rows), model_metrics.csv and feature_importances.csv exported from SQLite. A LA/NYC city slicer filters every visual, with cross-filtering enabled, so non-technical users can explore without background knowledge.

- KPI cards: total listings, price, distinct listing count and professional-host listings, all updating with the slicer.
- Overview row: average price by room type, listings by city, top features by model importance, and a dual-axis chart of booking activity vs superhost rate (visualising the paradox).
- Model row: donut charts comparing classifiers (F1) and regressors (R^2), superhost rate by city (LA ≈38.6% vs NYC ≈21.5%), and an actual vs predicted average price scatter by neighbourhood showing under-prediction at the high end.
- Time and composition row: availability over the 12-month forward calendar window and a 100% stacked bar of professional vs non-professional listings by city.

The layout follows one story: market size -> price drivers -> who becomes a superhost -> can price be predicted -> when to list. Design uses varied chart types, a consistent purple palette and business-language titles.



9. References

- Inside Airbnb. (n.d.). Get the data. https://insideairbnb.com/get-the-data/